Unit 1

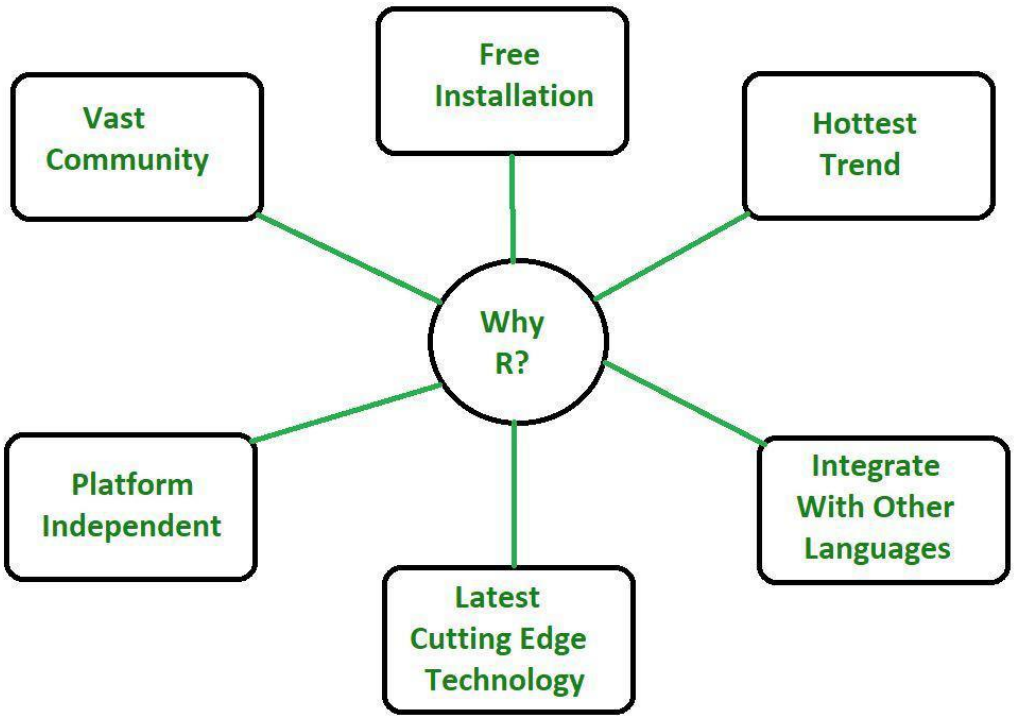
R Programming Language – Introduction

R is an open-source programming language that is widely used as a statistical software and data analysis tool.

R generally comes with the Command-line interface.

R is available across widely used platforms like Windows, Linux, and macOS. Also, the R programming language is the latest cutting-edge tool.

It was designed by **Ross Ihaka and Robert Gentleman** at the University of Auckland, New Zealand, and is currently being developed by the R Development Core Team.



What is R Programming Language?

- **R programming is used as a leading tool for machine learning, statistics, and data analysis. Objects, functions, and packages can easily be created by R.**
- **It's a platform-independent language**. This means it can be applied to all operating systems.

- **It's an open-source free language**. That means anyone can install it in any organization without purchasing a license.
- **R programming language is not only a statistic package but also allows us to integrate with other languages (C, C++).** Thus, you can easily interact with many data sources and statistical packages.
- The R programming language has a vast community of users and it's growing day by day.
- R is currently one of the most requested programming languages in the Data Science job market which makes it the hottest trend nowadays.

Why Use R?

- **Statistical Analysis:** R is designed for analysis and It provides an extensive collection of graphical and statistical techniques, By making a preferred choice for statisticians and data analysts.
- **Open Source:** R is an open – source software, which means it is freely available to anyone. It can be accessble by a vibrant community of users and developers.
- **Data Visulaization :** R boasts an array of libraries like ggplot2 that enable the creation of high-quality, customizable data visualizations.
- **Data Manipulation :** R offers tools that are for data manipulation and transformation. For example: IT simplifies the process of filtering , summarizing and transforming data.
- **Integration :** R can be easily integrate with other programming languages and data sources. IT has connectors to various databases and can be used in conjunction with python, SQL and other tools.
- **Community and Packages:** R has vast ecosystem of packages that extend its functionality. There are packages that can help you accomplish needs of analytics.

Features of R Programming Language

- **R Packages:** One of the major features of R is it has a wide availability of libraries. R has CRAN(**Comprehensive R Archive Network**), which is a repository holding more than 10, 0000 packages.

- **Distributed Computing:** Distributed computing is a model in which components of a software system are shared among multiple computers to improve efficiency and performance. Two new packages ddR and multidplyr used for distributed programming in R were released in November 2015.

Statistical Features of R

- **Basic Statistics:** The most common basic statistics terms are the mean, mode, and median. These are all known as "Measures of Central Tendency." So using the R language we can measure central tendency very easily.
- **Static graphics:** R is rich with facilities for creating and developing interesting static graphics. R contains functionality for many plot types including graphic maps, mosaic plots, biplots, and the list goes on.
- **Probability distributions:** Probability distributions play a vital role in statistics and by using R we can easily handle various types of probability distributions such as Binomial Distribution, Normal Distribution, Chi-squared Distribution, and many more.
- **Data analysis:** It provides a large, coherent, and integrated collection of tools for data analysis.

Advantages of R

- R is the most comprehensive statistical analysis package. As new technology and concepts often appear first in R.
- As R programming language is an open source. Thus, you can run R anywhere and at any time.
- R programming language is suitable for GNU/Linux and Windows operating systems.
- R programming is cross-platform and runs on any operating system.
- In R, everyone is welcome to provide new packages, bug fixes, and code enhancements.

Disadvantages of R

- In the R programming language, the standard of some packages is less than perfect.
- Although, R commands give little pressure on memory management. So R programming language may consume all available memory.
- In R basically, nobody to complain if something doesn't work.
- R programming language is much slower than other programming languages such as Python and MATLAB.

Applications of R

- We use R for Data Science. It gives us a broad variety of libraries related to statistics. It also provides the environment for statistical computing and design.
- R is used by many quantitative analysts as its programming tool. Thus, it helps in data importing and cleaning.
- R is the most prevalent language. So many data analysts and research programmers use it. Hence, it is used as a fundamental tool for finance.
- Tech giants like Google, Facebook, Bing, Twitter, Accenture, Wipro, and many more using R nowadays.

<u>Difference between R Programming and Python Programming</u>

Feature	**R**	**Python**
Introduction	R is a language and environment for statistical programming which includes statistical computing and graphics.	Python is a general-purpose programming language for data analysis and scientific computing

Feature	R	Python
Objective	It has many features which are useful for statistical analysis and representation.	It can be used to develop GUI applications and web applications as well as with embedded systems
Workability	It has many easy-to-use packages for performing tasks	It can easily perform matrix computation as well as optimization
Integrated development environment	Various popular R IDEs are Rstudio, RKward, R commander, etc.	Various popular Python IDEs are Spyder, Eclipse+Pydev, Atom, etc.
Libraries and packages	There are many packages and libraries like ggplot2, caret, etc.	Some essential packages and libraries are Pandas, Numpy, Scipy, etc.
Scope	It is mainly used for complex data analysis in data science.	It takes a more streamlined approach for data science projects.

Installation of R

R programming is a very popular language and to work on that we have to install two things, i.e., R and RStudio. R and RStudio works together to create a project on R.

Installing R to the local computer is very easy. First, we must know which operating system we are using so that we can download it accordingly.

The official site https://cloud.r-project.org provides binary files for major operating systems including Windows, Linux, and Mac OS. In some Linux distributions, R is installed by default, which we can verify from the console by entering R.

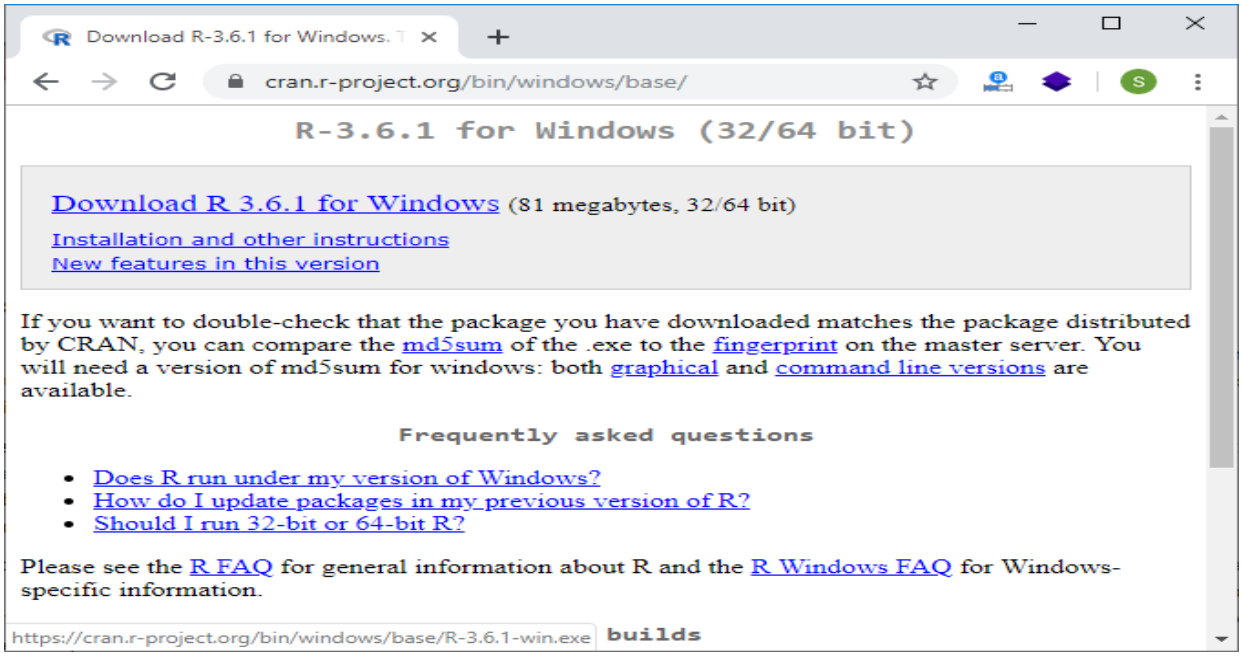
To install R, either we can get it from the site https://cloud.r-project.org or can use commands from the terminal.

Install R in Windows

There are following steps used to install the R in Windows:

Step 1:

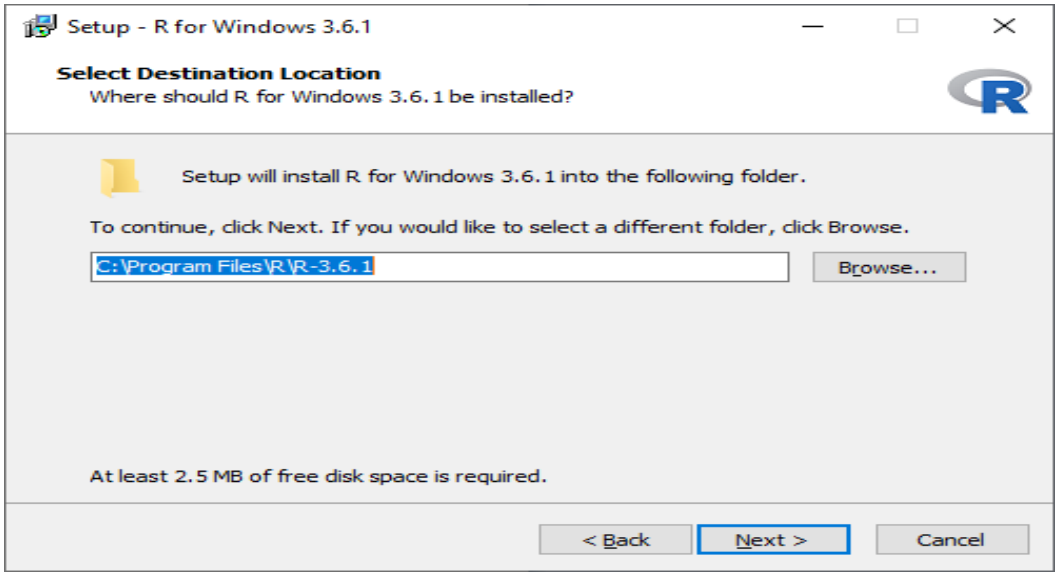
First, we have to download the R setup from https://cloud.r-project.org/bin/windows/base/.



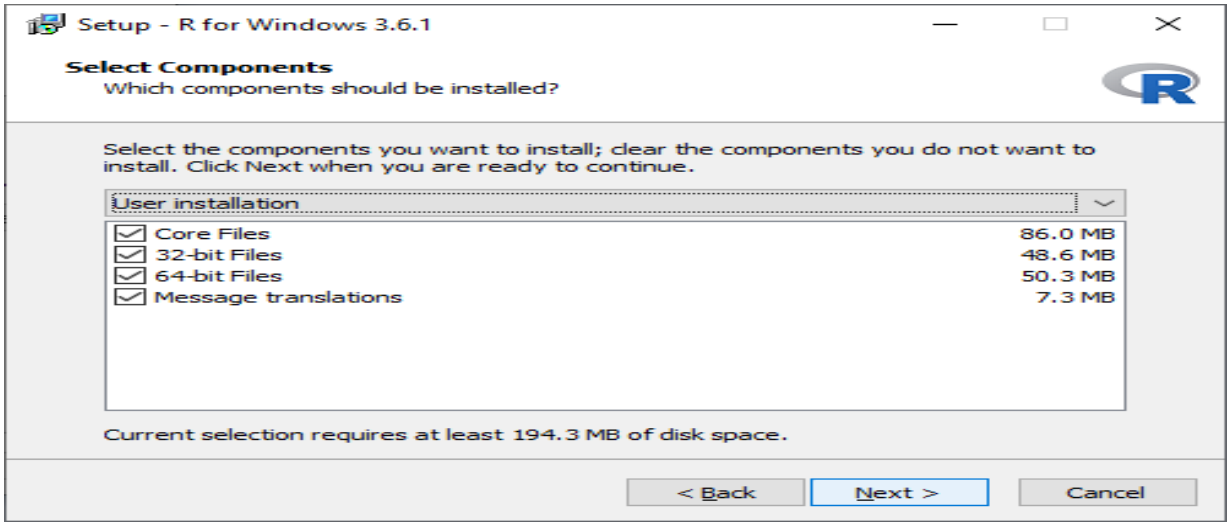
Step 2:

When we click on **Download R 3.6.1 for windows**, our downloading will be started of R setup. Once the downloading is finished, we have to run the setup of R in the following way:

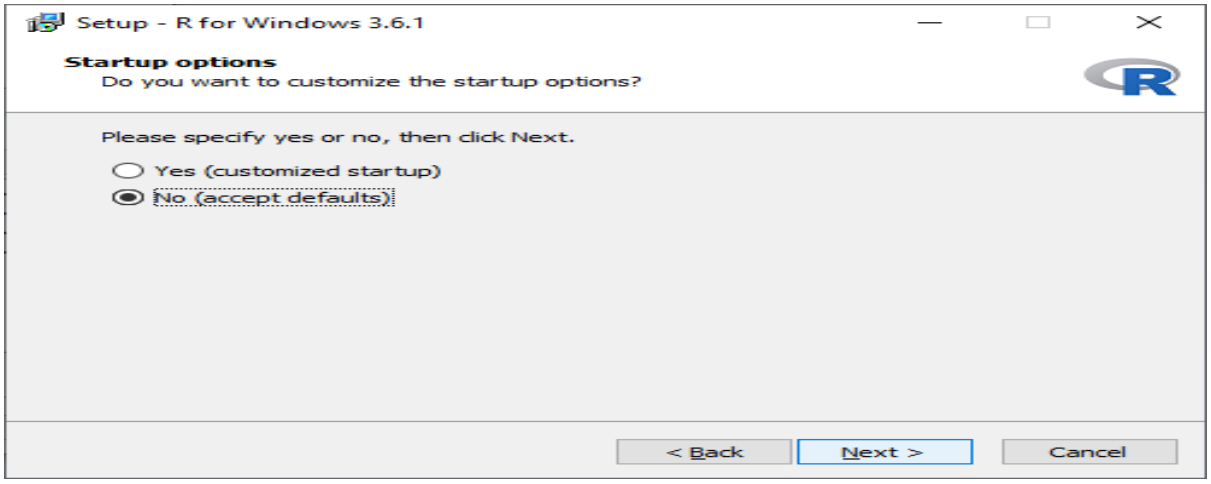
1) Select the path where we want to download the R and proceed to Next.



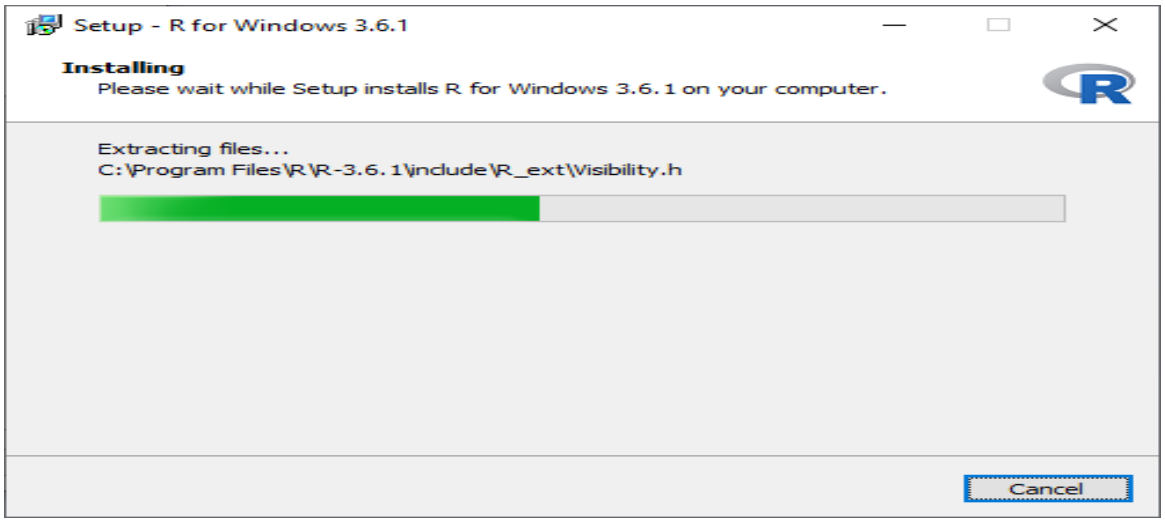
2) Select all components which we want to install, and then we will proceed to **Next**.



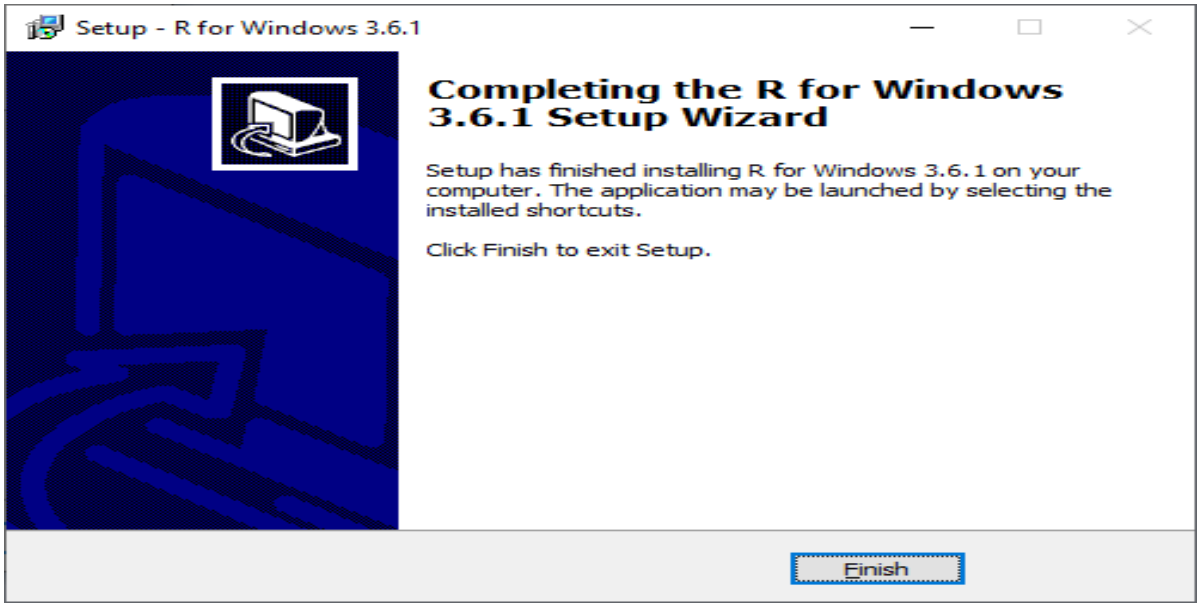
3) In the next step, we have to select either customized startup or accept the default, and then we proceed to **Next**.



4) When we proceed to next, our installation of R in our system will get started:



5) In the last, we will click on finish to successfully install R in our system.



Introduction to R Studio

R Studio is an integrated development environment (IDE) for R. IDE is a GUI.

RStudio is an open-source IDE (integrated development environment) that is widely used as a graphical front-end for working with the R programming language starting from version 3.0.1. It has many helpful features that make it very popular among R users:

- User-friendly
- Flexible
- Multifunctional
- Allows creating reusable scripts
- Tracks operational history
- Autocompletes the code
- Offers detailed and comprehensive help on any object
- Provides easy access to all imported data and built objects
- Makes it easy to switch between terminal and console
- Allows plot previewing
- Supports efficient project creation and sharing
- Can be used with other programming languages (Python, SQL, etc.)
- R Studio is available as both Open source and Commercial software.
- R Studio is also available as both Desktop and Server versions.
- R Studio is also available for various platforms such as Windows, Linux, and macOS.
- It is a powerful IDE, specifically used for the R language.
- Provides literate programming tools, which allow the use of R scripts, outputs, text, and images in reports, Word documents, and even HTML files.

- The use of Shiny (open-source R package) allows us to create interactive content in reports and presentations.

R Command Prompt

It is required that we have already installed the R environment set up in our system to work on the R command prompt. After the installation of R environment setup, we can easily start R command prompt by typing R in our Windows command prompt. When we press enter after typing R, it will launch interpreter, and we will get a prompt on which we can code our program.

```
Rterm (64-bit)
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\ajeet>R

R version 3.6.1 (2019-07-05) -- "Action of the Toes"
Copyright (C) 2019 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

"Hello, World!" Program

The code of "Hello World!" in R programming can be written as:

```
> string <- "Hello World!"
> print(string)
[1] "Hello World!"
>
```

In the above code, the first statement defines a **string variable** string, where we assign a string "Hello World!". The next statement print() is used to print the value which is stored in the variable string.

R Script File

The R script file is another way on which we can write our programs, and then we execute those scripts at our command prompt with the help of R interpreter known as **Rscript**. We make a text file and write the following code. We will save this file with .R extension as:

Demo.R

1. string <-"Hello World!"
2. **print**(string)

To execute this file in Windows and other operating systems, the process will remain the same as mentioned below.

```
Command Prompt

C:\Users\ajeet\R>Rscript Hello.R
```

When we press enter it will give us the following output:

```
[1] "Hello World!"
```

Comments

In R programming, comments are the programmer readable explanation in the source code of an R program. The purpose of adding these comments is to make the source code easier to understand. These comments are generally ignored by compilers and interpreters.

In R programming there is only single-line comment. R doesn't support multi-line comment. But if we want to perform multi-line comments, then we can add our code in a false block.

Single-line comment

```
#My First program in R programming
string <-"Hello World!"
print(string)
```

The trick for multi-line comment

```
#Trick for multi-line comment
if(FALSE) {
  "R is an interpreted computer programming language which was create
d by
```

Ross Ihaka **and** Robert Gentleman at the University of Auckland, New Z ealand "

}

#My First program in R programming

string <-"Hello World!"

print(string)

Update, Remove and Check Installed Packages in R

1) **To check what packages are installed on your computer, type this command:**

installed.packages()

2) **To update all the packages, type this command:**

update.packages()

3) **To update a specific package, type this command:**

install.packages("PACKAGE NAME")

4) TO returns path to the locations where the given packages are found.

find. Package()

5) package Description()

If a 'DESCRIPTION' file for the given package is found and can successfully be read, packageDescription returns an object of class "packageDescription" , which is a named list with the values of the (given) fields as elements and the tags as names, unless drop = TRUE .

6) help ()

help operator in R provide access to the documentation pages for R functions, data sets, and other objects, both for packages in the standard R distribution and for contributed packages.

Difference Between a Package and a Library

There is always confusion between a package and a library, and we find people calling libraries as packages.

- **library():** It is the command used to load a package, and it refers to the place where the package is contained, usually a folder on our computer.
- **Package:** It is a collection of functions bundled conveniently. The package is an appropriate way to organize our own work and share it with others.

Special Value Functions in R

Three additional values to represent unexpected results, namely the

1. NULL,
2. NaN,
3. Inf
4. -Inf

1) Undefined field of a list (NULL):

a) The first scenario of NULL is when you try to access an element of a list that is undefined.

```
my_list <- list(num = 1:3, char = c("a", "b"))

my_list$logi

#> NULL
```

Here, the result is NULL since logi is not a defined field in my_list.

b) ***Remove an element from a list***
c) ***Initialize a list of certain length***
d) ***NULL values when creating a vector***

2) NaN : NaN, represents **Not a Number**, usually appears when you divide 0 by 0.
0/0
#> [1] NaN
Again, it is worth to look at str(), typeof() and length() of NaN.

3) Inf : The last special we want to introduce in this section is Inf, representing the value is positive infinity (∞), corresponding to a proper mathematical limit. Similarly, we also have negative infinity: -Inf.

Example:

The first scenario of NULL is when you try to access an element of a list that is undefined.

A comparison of the four special values in R

summarize the different behaviours of the four special values in R in the following table.

Summary	NA	NULL	NaN	Inf
class()	"logical"	"NULL"	"numeric"	"numeric"
length()	1	0	1	1
check	is.na()	is.null()	is.nan()	is.finite()

Input and output functions:

Input and output functions are a programming language's **ability to talk and interact with its users. Have a look at many such functions in R.**

With R, we can read inputs from the user or a file using simple and easy-to-use functions. Similarly, we can display the complex output or store it to a file using the same. R's base package has many such functions, and there are packages that provide functions that can do the same and process the information in the required ways at the same time.

In R, there are multiple ways to read and save input given by the user. here are a few of them:

1. readline() function

We can read the input given by the user in the terminal with the `readline()` function.
Code:

```
input_read <- readline()
```

2. scan() function

We can also use the `scan()` function to read user input. This function, however, can only read numeric values and returns a numeric vector. If a non-numeric input is given, the function gives an error.

```
input_scan <- scan()
```

To display the output of your program to the screen, you can use one of the following functions:

1. print() functions

We can use the `print()` function to display the output to the terminal. The `print()` function is a generic function. This means that the function has a lot of different methods for different types of objects it may need to print. The function takes an object as the argument. For example:

```
print(input_read)
```

2. cat() function

We can also use the `cat()` function to display a string. The `cat()` function concatenates all of the arguments and forms a single string which it then prints. For example:
Code:

```
cat("hello", "this","is","techvidvan",12345,TRUE)
```

How to Write in a File?

The `write.table()` function is a very handy way to write tabular data to a file.

How to Read From a File?

Depending upon the data and the type of the file, there are multiple functions in R that can be used to read data from a file. they are:

1. scan() function

The `scan()` function can read data from a file as well. The data may be stored as a **data frame** and, therefore, may require conversions into desired formats.

2. readlines() function

We can also use the `readlines()` function to read a file one line at a time.

3. read.table() function

The `read.table()` function is another useful function that can be used to read data from a file and store it in the form of a table.

Step Install R on windows

Step 1 Go to **cran.r-project.org** website.

Step – 2: Click on the Download R for Windows link.

Step – 3: Click on the **base** subdirectory link or **install R for the first time** link.
Step – 4: Click **Download R X.X.X for Windows** (X.X.X stand for the latest version of R. eg: 3.6.1) and save the executable .exe file.

Step – 5: Run the **.exe** file and follow the installation instructions.

- 5.a. Select the desired language and then click **Next**.
- 5.b. Read the license agreement and click **Next**.
- 5.c. Select the components you wish to install (it is recommended to install all the components). Click **Next**.
- 5.d. Enter/browse the folder/path you wish to install R into and then confirm by clicking **Next**.
- 5.e. Select additional tasks like creating desktop shortcuts etc. then click **Next**.
- 5.f. Wait for the installation process to complete.
- 5.g. Click on **Finish** to complete the installation.